

COMPOSABLE GEOMETRY — THE MERGE CHAIN

One depth estimate, four derivation strategies, combined like Nuke Merge nodes.

FILE `atlas_composable_geometry_templecity_workflow.json` VERIFIED 2026-07-16

THE STORY

Some scenes don't fit one preset. This AI temple-city has vertical walls, genuine spires (which plain wall-fitting truncates at the first sloped roof), sloped facades, AND organic terrain — so the workflow estimates depth ONCE (AtlasDepthMap carries the raw metric result by reference) and fans it into four strategies: Walls with 3 distance modes, TowersSpires with roofline splitting, RoofsFacades via sequential RANSAC, and a relief mesh.

The crux is the combination. Chaining derive nodes directly would silently erase each other's geometry — every derive node strips prior proxy geometry by design, so re-running never piles up stale meshes. AtlasMergeGeometry is the one explicit place branches combine, exactly like a Nuke Merge: binary, chainable, solve_a's camera wins, backdrop deduped. Three merges fold the four branches into one scene, which exports as a single-projection Nuke .nk carrying the real relief OBJ.

IN THE VIEWPORT — PROJECTED VS. THE GREY MESH



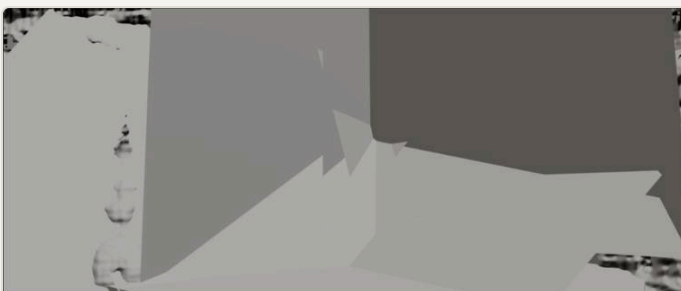
SOURCE PLATE

atlas_00022_templecity.png · AI vista · 9216×3840 — the single still photograph everything below is reconstructed from.



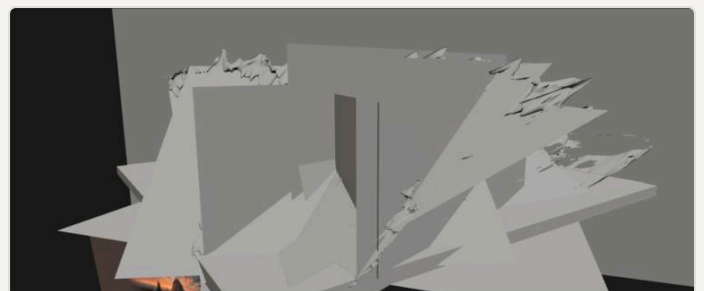
PROJECT ON

The photo reassembled on the recovered camera — texels assigned by ray, perfect from this view.



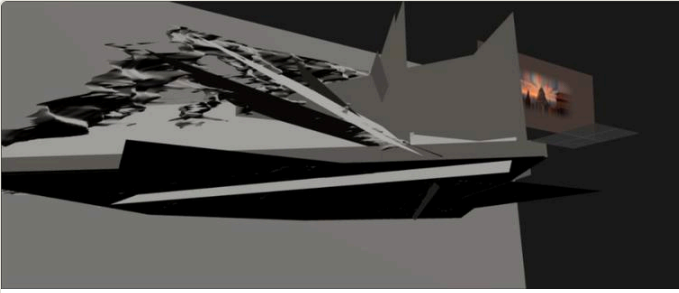
PROJECT OFF

The same geometry as a grey mesh, recovered view — what the depth-derived surfaces actually are.



GREY · ORBIT LEFT

Orbited off-axis: the reconstructed cone. Coverage ends where the camera never saw.



GREY · ORBIT RIGHT

The opposite angle — silhouette tears read as deliberate holes, not errors.

NODES ON STAGE

AtlasDepthMap

AtlasDeriveWalls

AtlasDeriveTowersSpires

AtlasDeriveRoofsFacades

AtlasDeriveReliefMesh

AtlasMergeGeometry ×3

AtlasScaleOverride

AtlasExportNuke

RETUNE ON YOUR PLATE

-  **camera_height_m = 16.** All vistas solve to the assumed 1.6 m default — ~10× small for this elevated view. Same dial as the NYC builds; calibrate per plate.
- **Per-branch knobs.** distance_modes on Walls, roofline_split on TowersSpires, max_planes on RoofsFacades. Bypass any branch — the merges combine whatever survives.
- **Merge order matters.** solve_a's camera wins; keep the relief branch first so its full-frame coverage is the base.
- **Known cosmetic.** Merges dedupe the backdrop but not projection_ground — a 3-way merge carries coincident ground planes. Harmless: identical texels by ray.

RUN-VERIFIED

RUN ✓ 18 merged primitives spanning all four strategies; the .nk carries Camera2/Project3D2/ReadGeo2 + the relief OBJ, drag-and-drop ready.

REQUIREMENTS

Atlas + [neural] · no gates

